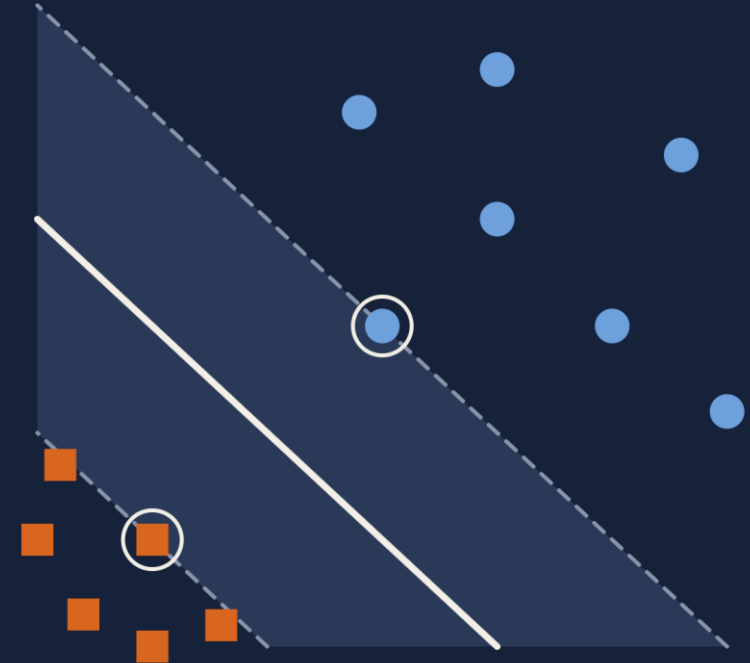


CLASSIFICATION METHODS

Support Vector Machines

Optimal separating hyperplane · Kernel functions

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1 • THE PROBLEM

Two classes, many possible dividing lines

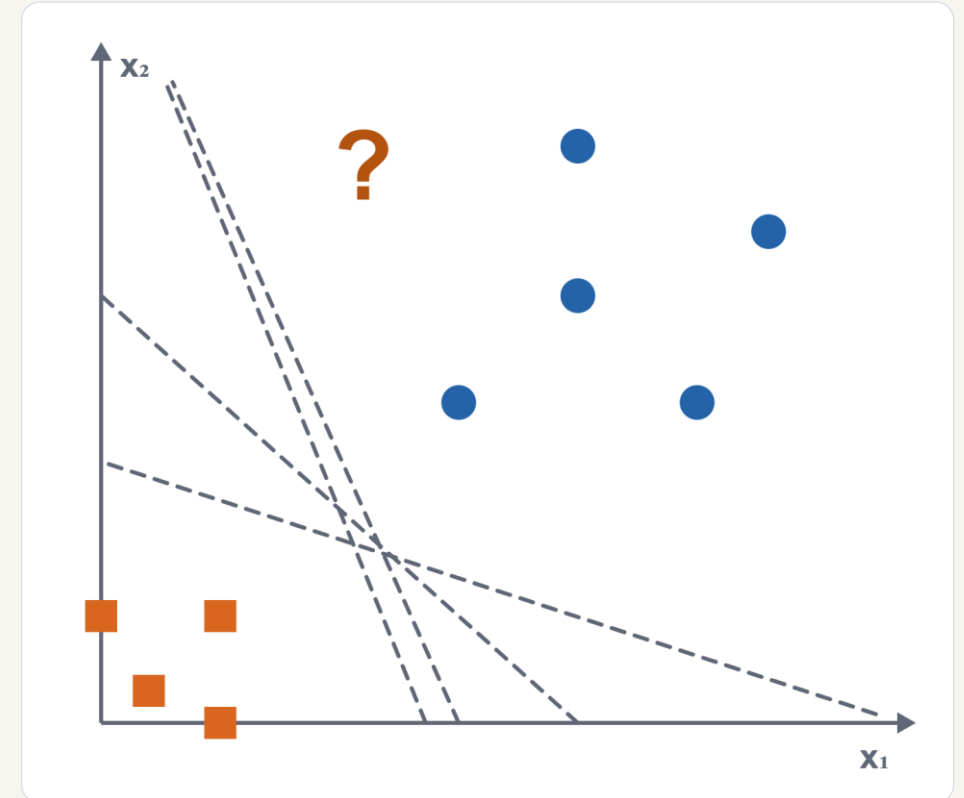
A classifier must draw a **boundary** between labelled points of two classes:

● class +1 and

■ class -1.

All four dashed lines separate the training data perfectly. But a line that passes **very close** to a point will misclassify a new point that lands just a little off.

SVM's answer: choose the line that stays as **far as possible** from both classes.



2 · DEFINITION

What is a Support Vector Machine?

A **Support Vector Machine (SVM)** is a supervised learning that classifies data by finding the **hyperplane** that separates the classes with the **maximum margin**.

Supervised

Learns from labelled examples (x_i, y_i) with $y_i \in \{+1, -1\}$.

Margin-based

Chooses the widest gap between classes, not just any separator.

Linear and non-linear

Straight boundaries directly; curved ones through kernel functions.

Basic SVM = binary classification

Multiclass SVM = combination of multiple binary SVMs

For example, with three classes A, B, and C, one common method is **One-vs-Rest**:

- SVM 1: A vs not-A
- SVM 2: B vs not-B
- SVM 3: C vs not-C

Then the outputs are compared and the class with the strongest score is selected.

Another method is **One-vs-One**:

- A vs B
- A vs C
- B vs C

Each classifier votes, and the class with the most votes is chosen.

2 · TERMINOLOGY

Term	Meaning	Symbol / formula
Hyperplane	Separating boundary: a line in 2-D, a plane in 3-D	$w \cdot x + b = 0$
Weight vector w	Perpendicular to the hyperplane; sets its direction	w
Bias b	Shifts the hyperplane away from the origin	b
Margin	Gap between the nearest points of the two classes	$2 / w $
Support vectors	Training points lying exactly on the margin lines	$y_i(w \cdot x_i + b) = 1$
Kernel function	Dot product in a higher-dimensional space	$K(x_i, x_j) = \varphi(x_i) \cdot \varphi(x_j)$
Soft margin, C	Allows some errors; C is the penalty per error	$C > 0$

3 · HYPERPLANE

The decision boundary: $w \cdot x + b = 0$

A **hyperplane** simply means a **flat boundary used to divide a space into two parts**.

A **hyperplane** in n dimensions is a flat boundary of $n - 1$ dimensions.

In **2D**: hyperplane has dimension $(2-1=1) \rightarrow$ a **line**

In **3D**: hyperplane has dimension $(3-1=2) \rightarrow$ a **plane**

In **4D**: hyperplane has dimension $(4-1=3) \rightarrow$ a 3D flat boundary inside 4D space

$$f(x) = w \cdot x + b$$

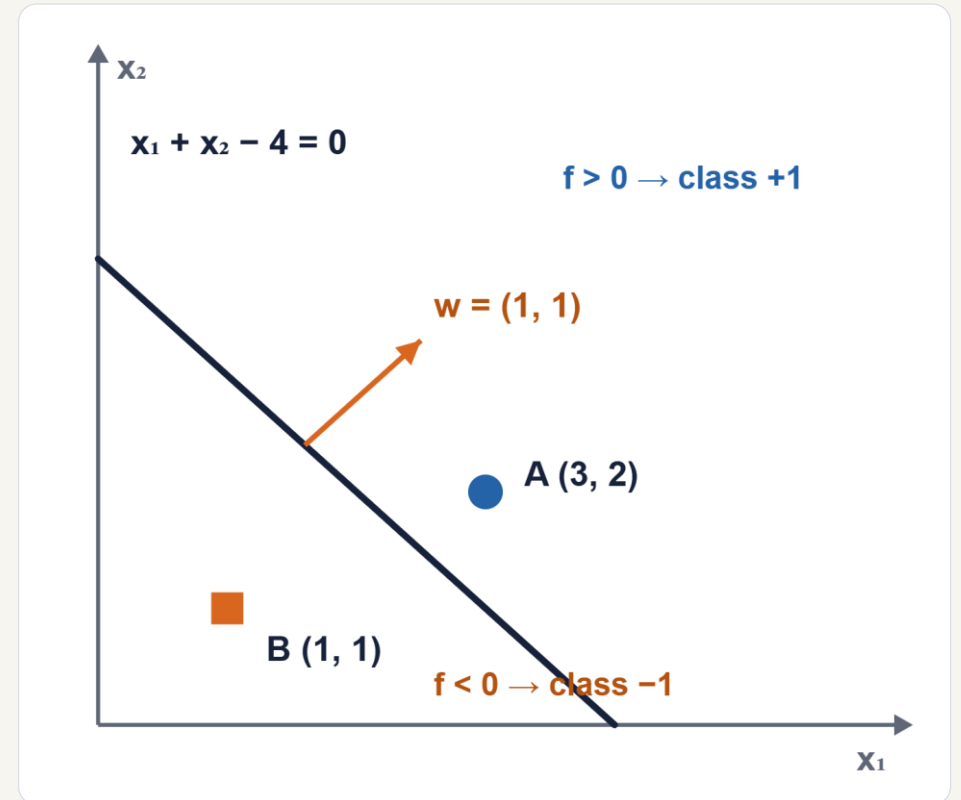
$$f(x) > 0 \rightarrow \text{class } +1 \quad f(x) < 0 \rightarrow \text{class } -1$$

EXAMPLE

$w = (1, 1)$, $b = -4$, so the line is $x_1 + x_2 = 4$.

$A(3, 2)$: $f = (3+2) - 4 = +1 \rightarrow \text{class } +1$

$B(1, 1)$: $f = (1+1) - 4 = -2 \rightarrow \text{class } -1$



4 • MARGIN

The margin is the gap the boundary leaves on each side

Draw two lines parallel to the hyperplane, through the nearest point of each class:

For points on one side, $w \cdot x + b > 0$
and on the other side, $w \cdot x + b < 0$

$$w \cdot x + b = +1$$

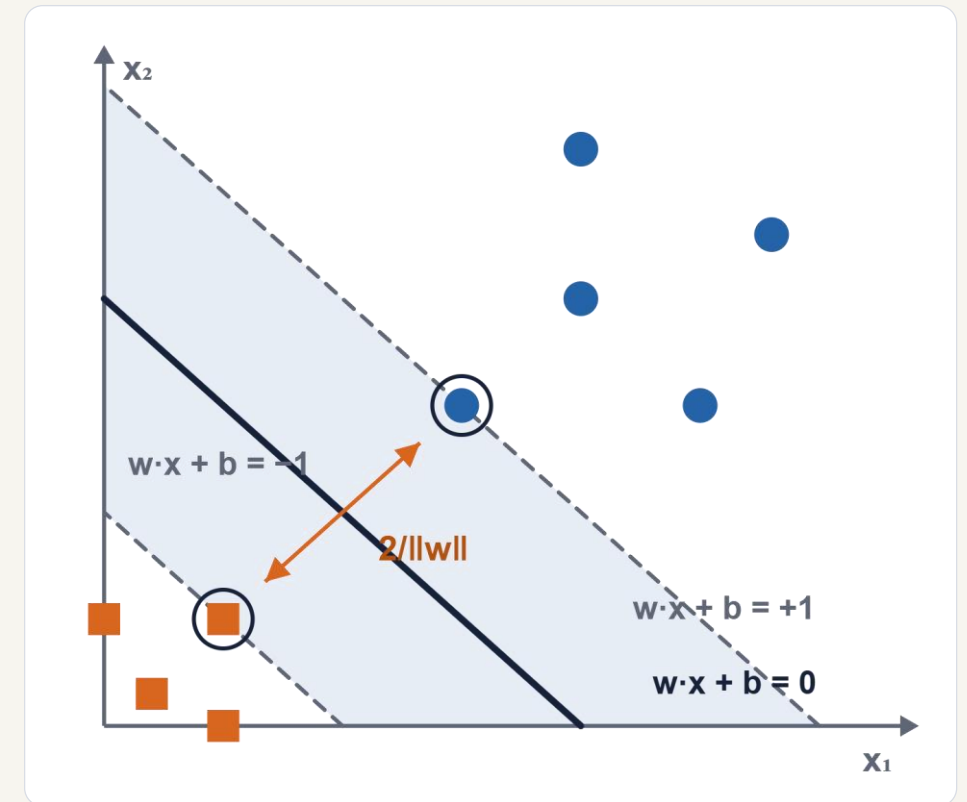
$$w \cdot x + b = -1$$

+1 and -1 are chosen by normalization;
they are not naturally fixed values

The distance between them is the **margin**:

$$\text{margin} = 2 / \|w\|$$

A wider margin leaves more room for noise, so new points are less likely to fall on the wrong side.



4 · OPTIMAL SEPARATING HYPERPLANE

Of all separators, pick the one with the widest margin

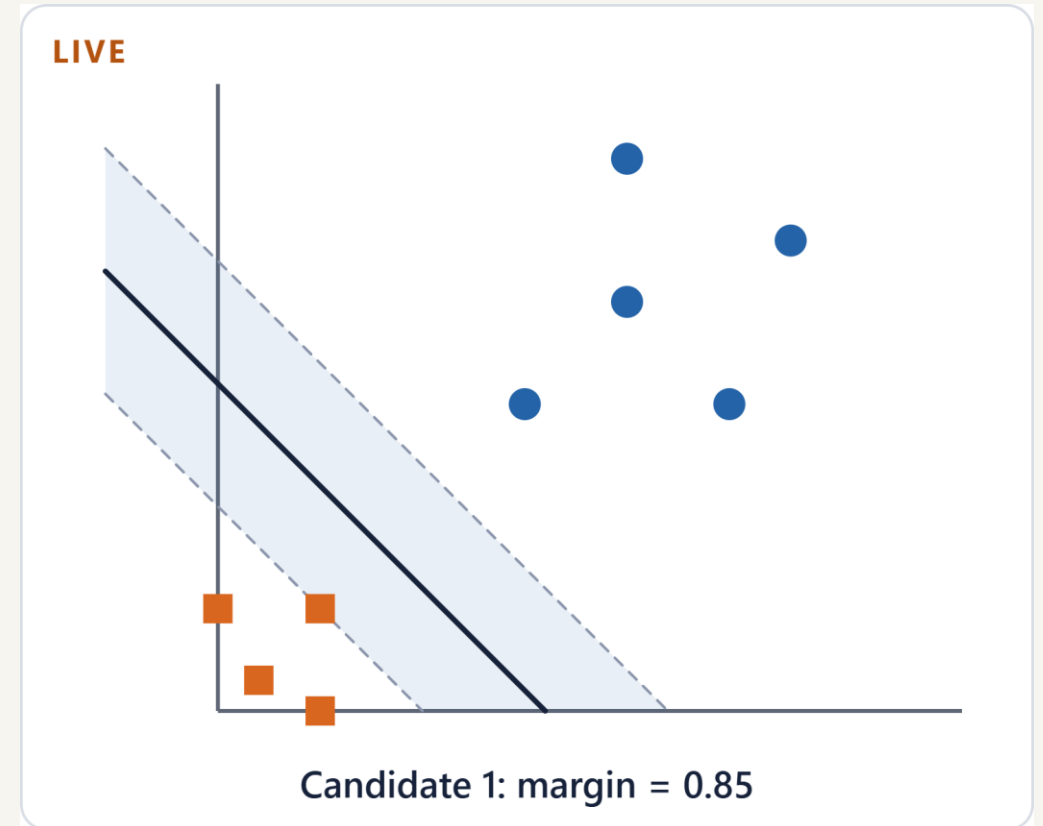
The **optimal separating hyperplane** is the separating hyperplane whose margin is **maximum**. Maximising $2/||w||$ is the same as minimising $||w||$:

$$\text{minimise } \frac{1}{2}||w||^2$$

$$\text{subject to } y_i(w \cdot x_i + b) \geq 1 \text{ for all } i$$

Why it matters

- Unique best answer (a convex problem, one global optimum)
- Most tolerant of noise near the boundary
- Generalises better to unseen data
- Depends only on the support vectors



Why SVM Minimizes $\|w\|^2$, Not $\|w\|$

1

The Magnitude of the Weight Vector

$$\|w\| = \sqrt{w_1^2 + w_2^2 + \dots + w_n^2}$$

The **square root** makes differentiation messier — its derivative brings in an extra $1/(2\|w\|)$ term that complicates the optimization.

2

SVM Minimizes the Square Instead

$$\|w\|^2 = w_1^2 + w_2^2 + \dots + w_n^2$$

Dropping the square root gives a clean polynomial — *much easier to differentiate*, and it reaches its minimum at the exact same w as the original.

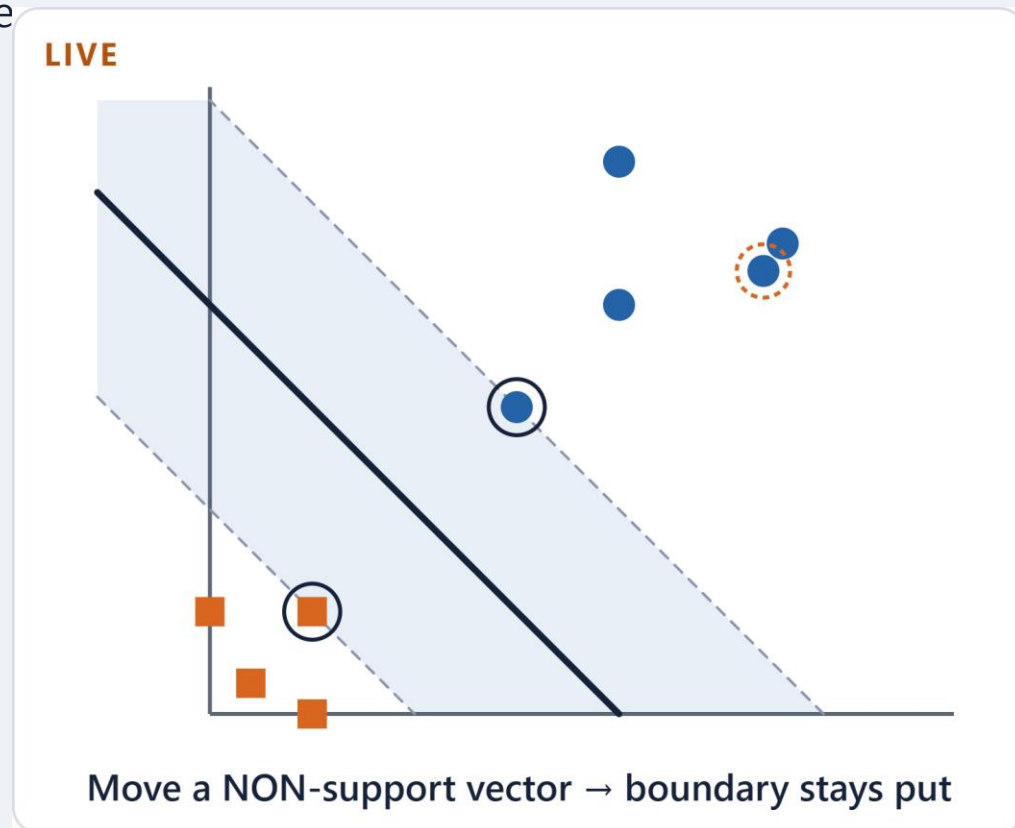
5 · SUPPORT VECTORS

The few points that hold the boundary in place

Support vectors are the training points closest to the hyperplane. They lie exactly on the margin lines, where $y_i(w \cdot x_i + b) = 1$.
 y_i is the class

Their role in classification

- **They alone define** the hyperplane: $w = \sum \alpha_i y_i x_i$, and $\alpha_i > 0$ only for support vectors
- Moving or removing any other point **changes nothing**
- Moving a support vector **shifts the boundary**
- Prediction needs only them, so the model is **compact**
- They are the hardest, most informative cases



Why SVM is called a maximum-margin classifier

Among all hyperplanes that separate the training data, SVM **explicitly chooses the one that maximises the margin**: the distance to the nearest points of both classes.

The margin is the objective

SVM's optimisation maximises $2/\|w\|$ (minimises $\frac{1}{2} \|w\|^2$).

Other classifiers may stop at any separator.

A safety buffer

New points that fall slightly off the training data still land on the correct side.

Better generalisation

A wider margin means a simpler, more stable boundary and lower error on unseen data.

[SVM Interactive Web Page](#)

5 · SOFT MARGIN

Real data overlaps, so allow a few mistakes

Slack variables ξ_i let some points sit inside the margin or on the wrong side. C controls how much each violation costs:

minimise $\frac{1}{2} \|w\|^2 + C \sum \xi_i$ • ξ_i = how much the i -th point violates the margin
• C = how strongly we punish those violations

Large C

Violations are expensive → narrow margin, few training errors.
Risk: overfitting to noise.

Small C

Violations are cheap → wider margin, some training errors. Usually generalises better.

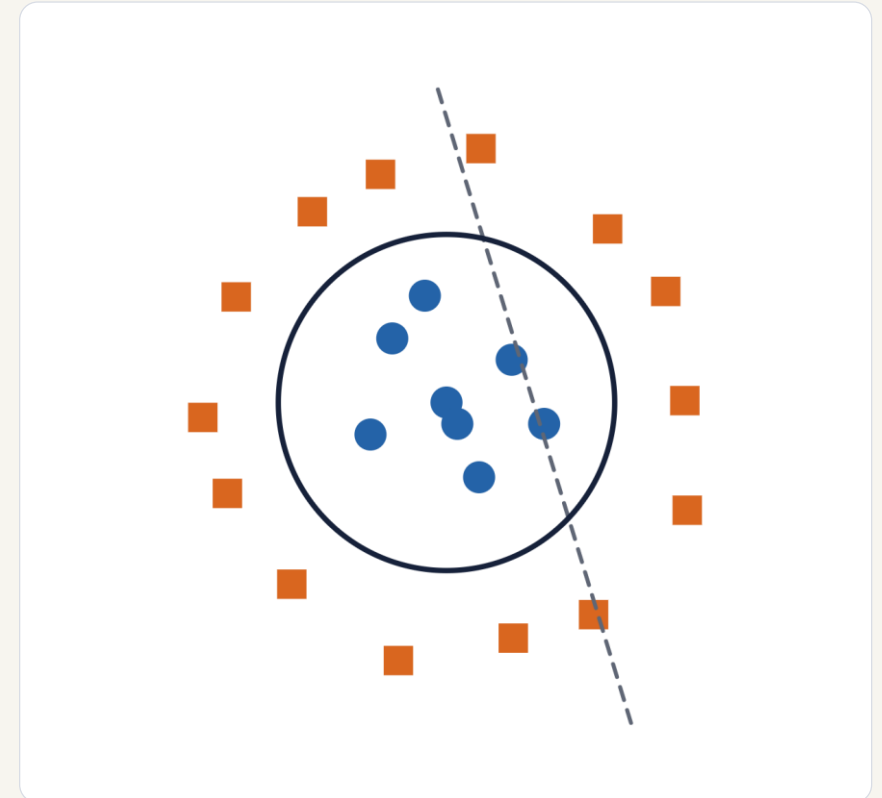
6 · NON-LINEAR DATA

When no straight line can separate the classes

Here class +1 sits **inside a ring** of class -1. Any straight line (dashed) misclassifies many points. The true boundary is a **circle**.

Idea: move the data into a higher-dimensional space where a straight hyperplane **does** separate it.

Example: add a new feature $z = x_1^2 + x_2^2$.
Inner points get small z , ring points large z .
A flat plane $z = \text{constant}$ now separates them.



6 · MAPPING TO A HIGHER DIMENSION

Lift the data, and a straight line works

On a single axis x , orange points sit at both ends and blue in the middle. **No single cut** separates them.

$$\varphi(x) = (x, x^2)$$

After the mapping, the classes are split by the **straight line** $x^2 = 4$.
Back on the original axis, that is the non-linear rule $|x| < 2$.

A **kernel function** lets SVM use such mappings without ever computing $\phi(x)$ explicitly.

LIVE



6 · KERNEL FUNCTION

Dot products in a high-dimensional space, computed cheaply

A **kernel function** $K(x_i, x_j)$ returns the dot product of two points **after** mapping them to a higher-dimensional space, without computing that mapping

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

SVM's training and prediction only ever need dot products, so replacing $x \cdot z$ with $K(x, z)$ is enough. This is the **kernel trick**.

EXAMPLE · $K(x, z) = (x \cdot z)^2$

$$x = (1, 2), z = (3, 4)$$

Kernel route: $x \cdot z = 1 \cdot 3 + 2 \cdot 4 = 11$, so $K = 11^2 = 121$

Mapping route: $\phi(x) = (x_1^2, \sqrt{2} \cdot x_1 x_2, x_2^2)$

$$\phi(x) = (1, 2\sqrt{2}, 4), \phi(z) = (9, 12\sqrt{2}, 16)$$

$$\phi(x) \cdot \phi(z) = 9 + 48 + 64 = 121 \checkmark$$

Same answer, but the kernel never built the 3-D vectors.

6 · COMMON KERNELS

Four kernels you should know

Kernel	Formula	Use when
Linear	$K(x, z) = x \cdot z$	Data is (nearly) linearly separable; many features, e.g. text
Polynomial	$K(x, z) = (x \cdot z + c)^d$	Curved boundaries of degree d ; feature interactions
RBF (Gaussian)	$K(x, z) = \exp(-\gamma \ x - z\ ^2)$	Default choice; complex, circular or irregular boundaries
Sigmoid	$K(x, z) = \tanh(\kappa x \cdot z + c)$	Behaves like a simple neural network

γ , c , d and κ are parameters chosen by the user (usually by cross-validation). A valid kernel must satisfy **Mercer's condition**: its matrix of values is symmetric and positive semi-definite.

What kernels make possible

Non-linear boundaries

A linear algorithm draws curved boundaries in the original space.

Cheap computation

No need to compute $\phi(x)$, even for infinite-dimensional spaces (RBF).

Flexibility

Pick the kernel that suits the data; special kernels exist for text, strings and graphs.

Same optimisation

The problem stays convex, so a single global optimum is still found.

Only dot products

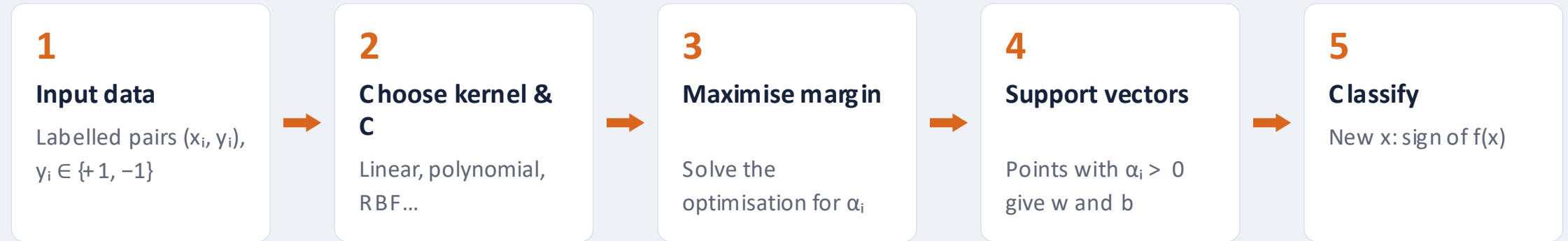
Decision function uses K alone: $f(x) = \text{sign}(\sum \alpha_i y_i K(x_i, x) + b)$.

Similarity measure

$K(x, z)$ says how alike two points are; RBF is 1 for identical points, near 0 for distant ones.

7 · WORKING OF SVM

How SVM performs binary classification, step by step



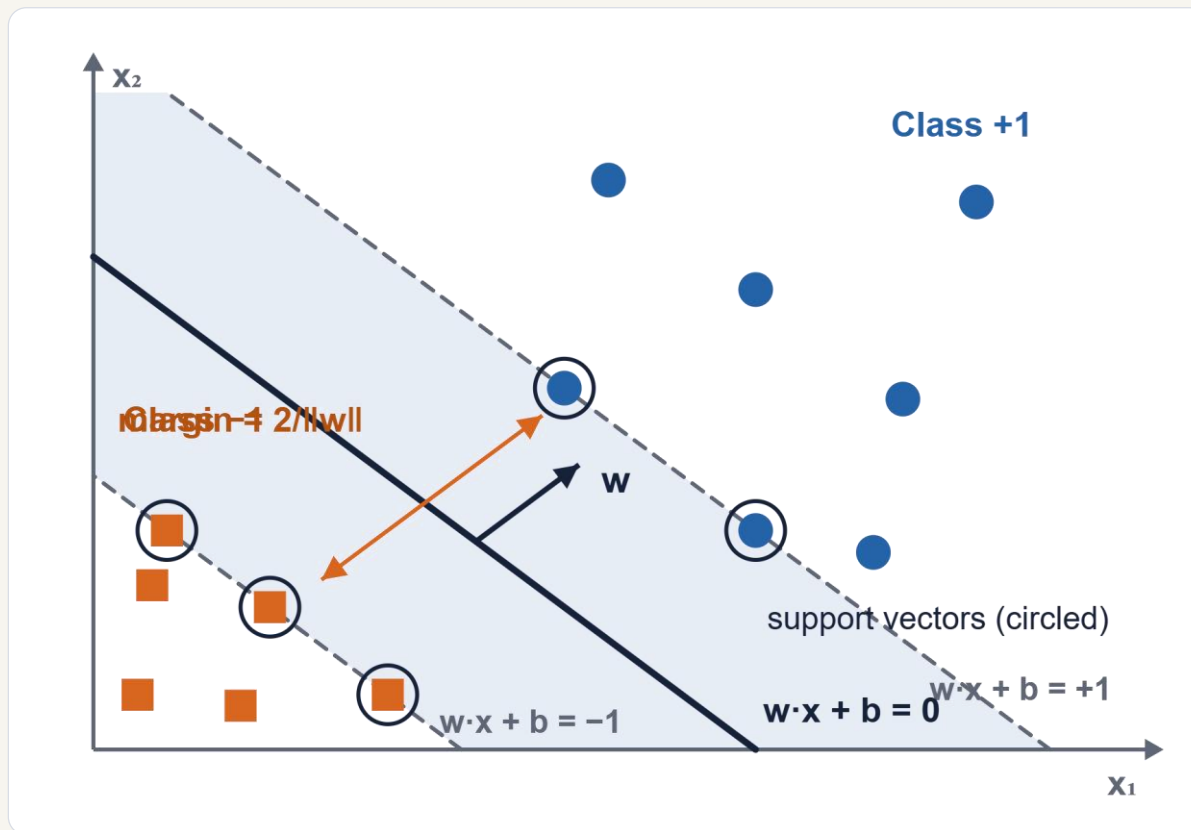
DECISION FUNCTION

$$f(x) = \text{sign}(\sum \alpha_i y_i K(x_i, x) + b)$$

The sum runs over support vectors only. For a linear kernel this reduces to $\text{sign}(w \cdot x + b)$.

7 · THE SVM DIAGRAM

Binary classification with a maximum-margin hyperplane



To draw it

1. Axes x_1 , x_2 and two groups of points
2. Solid hyperplane $w \cdot x + b = 0$ in the gap
3. Dashed lines $w \cdot x + b = \pm 1$ through the nearest points
4. Circle the support vectors
5. Mark margin $2/\|w\|$ and the vector w

7 · ADVANTAGES & LIMITATIONS

Where SVM shines, and where it struggles

High dimensions

Works well even when features outnumber samples.

Memory efficient

Stores and uses only the support vectors.

Versatile

Kernels handle linear and non-linear boundaries.

Resists overfitting

Maximum margin and C control complexity.

Global optimum

Convex problem: no local minima.

Clear margins

Very accurate when classes are well separated.

Limitations: slow on very large datasets · kernel and C must be tuned · no direct probability output · needs feature scaling · weaker when classes overlap heavily

8 · PRACTICE 1

Classify points with a trained SVM

A trained linear SVM has $w = (2, -1)$ and $b = -1$.

$$f(x) = 2x_1 - x_2 - 1$$

Classify each point and say whether it lies on the boundary or on a margin line:

- A (2, 1)
- B (0, 3)
- C (1, 1)
- D (1, 0)

Click / press → in Present mode to reveal each answer.

$$\text{A: } f = 4 - 1 - 1 = 2 \rightarrow \text{class } +1$$

$$\text{B: } f = 0 - 3 - 1 = -4 \rightarrow \text{class } -1$$

$$\text{C: } f = 2 - 1 - 1 = 0 \rightarrow \text{on the hyperplane (undecided)}$$

$$\text{D: } f = 2 - 0 - 1 = +1 \rightarrow \text{class } +1, \text{ on the margin line: a support-vector position}$$

8 · PRACTICE 3

Choose the right kernel

Scenario A

Spam filter: 10,000 word-count features, classes almost linearly separable.

Scenario B

Class A points form a disc; class B forms a ring around it.

Scenario C

Only products of features up to degree 3 matter.

Name a kernel for each and give a one-line reason.

A → Linear. Already separable in high dimensions; fast.

B → RBF (or polynomial, $d = 2$). Boundary is a circle.

C → Polynomial, $d = 3$. It models degree-3 interactions.

Find the separating line with the largest possible margin.

Class +1: (3,3), (4,4), (5,3)

Class -1: (1,1), (0,1), (1,0)

Find the nearest opposite-class points

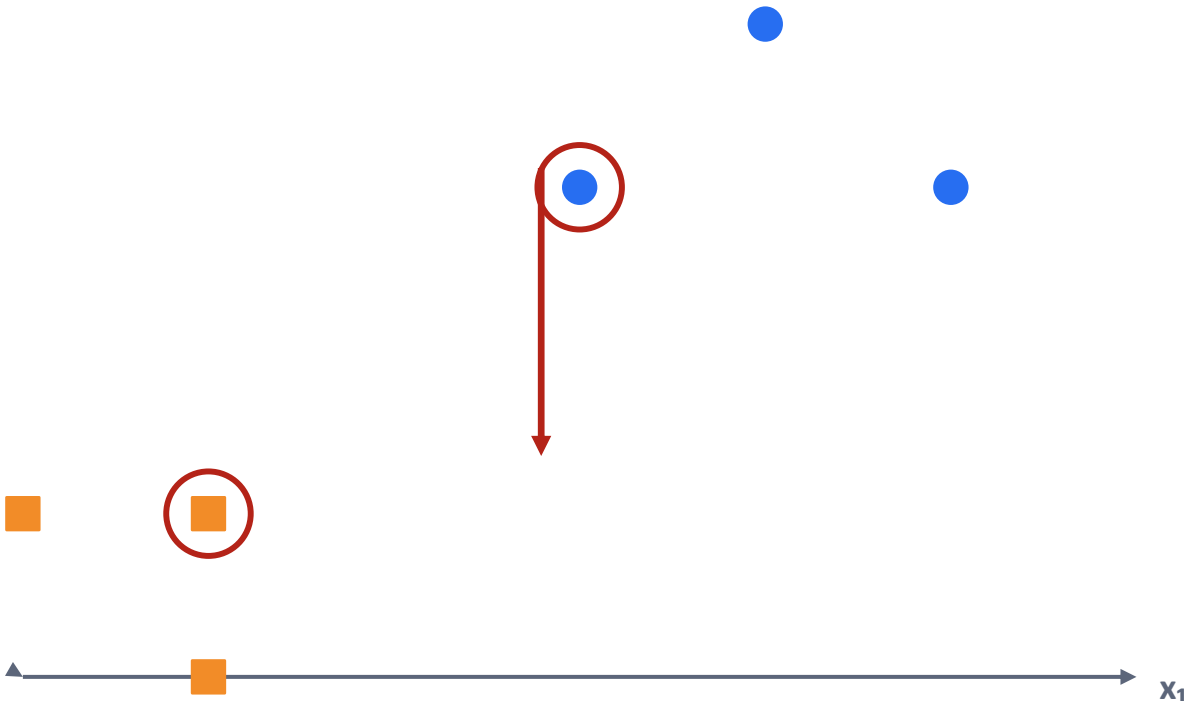
These nearest points become the support vectors.

x_2

Closest +1 point: (3,3)

Closest -1 point: (1,1)

These two points “hold” the margin.
Changing them can change the SVM
boundary.



Place the decision boundary halfway between them

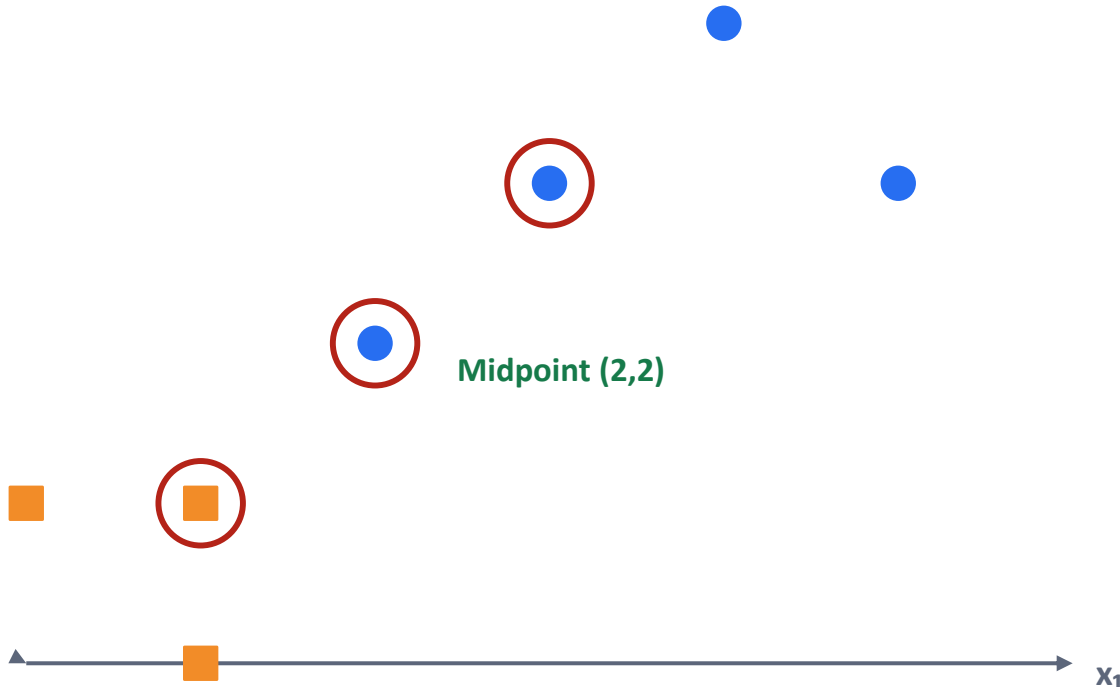
The best separating line must pass through the midpoint.

x_2

$$\text{Midpoint} = ((3+1)/2, (3+1)/2)$$

$$= (2, 2)$$

So the optimal hyperplane must pass through (2,2).



Find the slope of the separating line

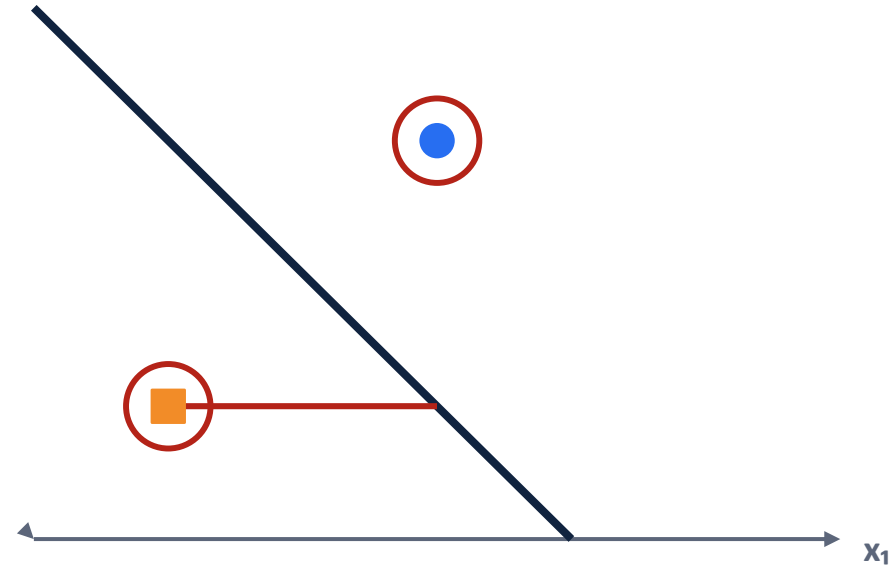
The decision boundary is perpendicular to the line joining support vectors.

Slope of line through $(1,1)$ and $(3,3)$

$$m = (3 - 1) / (3 - 1) = 1$$

Perpendicular slope = $-1 / m = -1$

x_2



Derive the optimal hyperplane equation

Use point-slope form with midpoint (2,2) and slope -1.

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -1(x - 2)$$

$$y - 2 = -x + 2$$

$$x + y - 4 = 0$$

Using feature names: $x_1 + x_2 - 4 = 0$
Equivalent form: $x_1 + x_2 = 4$

This is the optimal decision boundary.

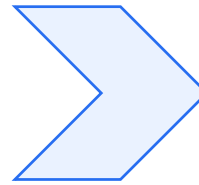
Convert to SVM standard scaling

The unscaled line works, but the SVM-normalized form gives +1 and -1 on support vectors.

$$\text{Unscaled: } \mathbf{x}_1 + \mathbf{x}_2 - 4 = 0$$

$$(3,3) \rightarrow +2 \quad \text{and} \quad (1,1) \rightarrow -2$$

Correct boundary, but not standard SVM scaling.



$\div 2$

$$\text{SVM form: } 0.5\mathbf{x}_1 + 0.5\mathbf{x}_2 - 2 = 0$$

$$(3,3) \rightarrow +1 \quad \text{and} \quad (1,1) \rightarrow -1$$

$$\mathbf{w} = [0.5, 0.5]^T, \quad \mathbf{b} = -2$$

Both are the same line. The second form is easier for SVM margin calculations.

Write the two margin boundaries

Now the support vectors exactly touch +1 and -1 margins.

Positive support vector (3,3)

$$0.5(3)+0.5(3)-2 = +1$$

$$0.5x_1 + 0.5x_2 - 2 = +1$$

$$x_1 + x_2 = 6$$

Negative support vector (1,1)

$$0.5(1)+0.5(1)-2 = -1$$

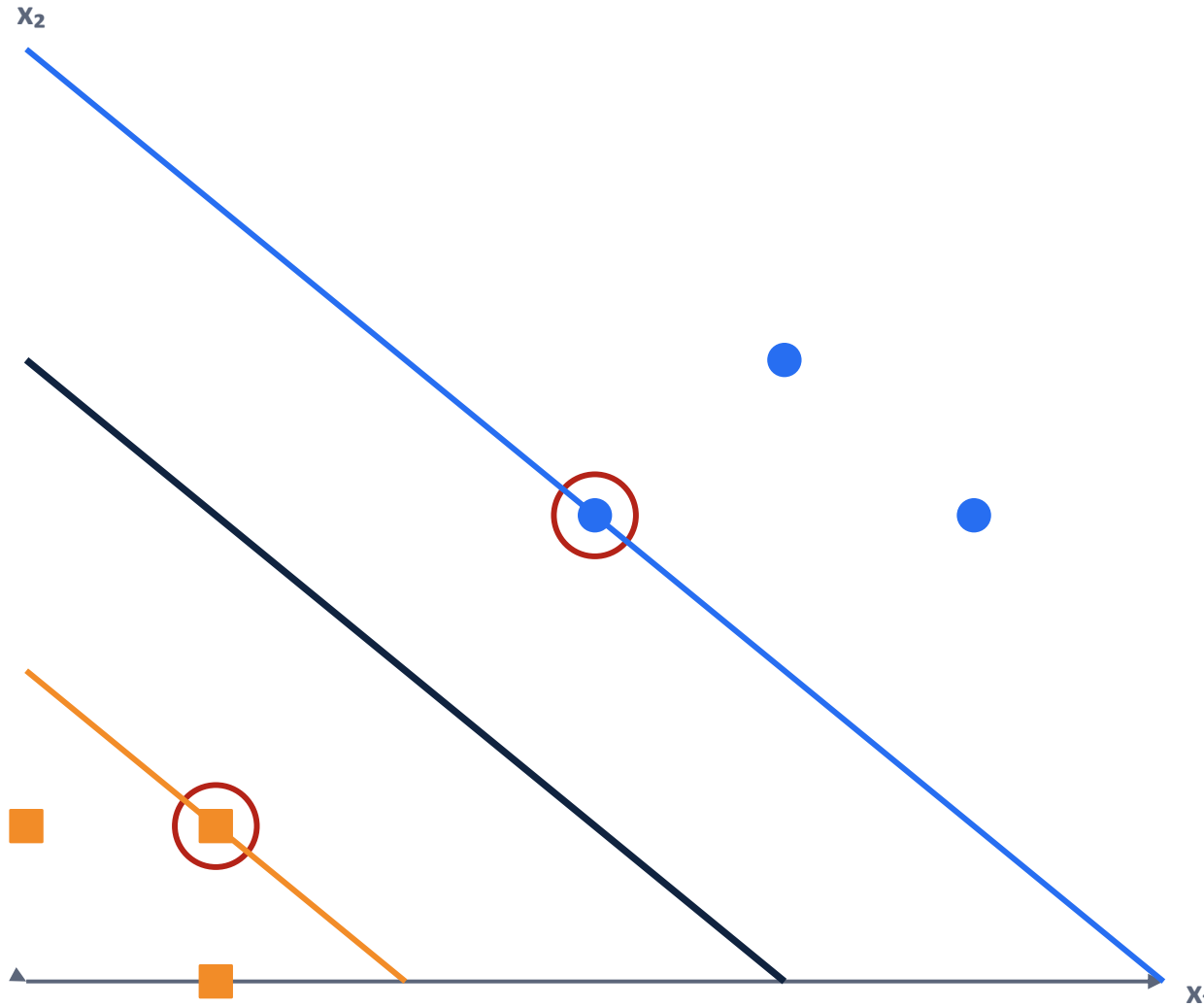
$$0.5x_1 + 0.5x_2 - 2 = -1$$

$$x_1 + x_2 = 2$$

Three parallel lines: margin, decision boundary, margin.

Visual view of the final result

The support vectors lie on the margin boundaries.



Positive margin in: $x_1+x_2 = 6$

Decision boundary: $x_1+x_2 = 4$

Negative margin in: $x_1+x_2 = 2$

Support vectors: (3,3) and (1,1)

Calculate the margin width

The total distance between the two margin lines is $2 / ||w||$.

$$w = [0.5, 0.5]^T$$

$$||w|| = \sqrt{(0.5)^2 + (0.5)^2}$$

$$= \sqrt{0.25 + 0.25} = \sqrt{0.5} = 1/\sqrt{2}$$

$$\text{Total margin} = 2 / ||w||$$

$$= 2 / (1/\sqrt{2}) = 2\sqrt{2}$$

$$\approx 2.83$$

$$\text{Distance from decision boundary to one margin} = 1 / ||w|| = \sqrt{2} \approx 1.414$$

Why minimize $\frac{1}{2} ||w||^2$?

SVM wants the margin to be as large as possible.

$$\text{Margin} = 2 / ||w||$$

To maximize the margin, we need to make $||w||$ small.

$$||w|| = \sqrt{w_1^2 + w_2^2 + \dots + w_n^2}$$

The square root makes differentiation less convenient.

$$||w||^2 = w_1^2 + w_2^2 + \dots + w_n^2$$

This is easier to differentiate.

$$\text{minimize } \frac{1}{2} ||w||^2$$

The $\frac{1}{2}$ only cancels the 2 during differentiation.

Why support vectors are important

Only the closest, most difficult points define the final boundary.

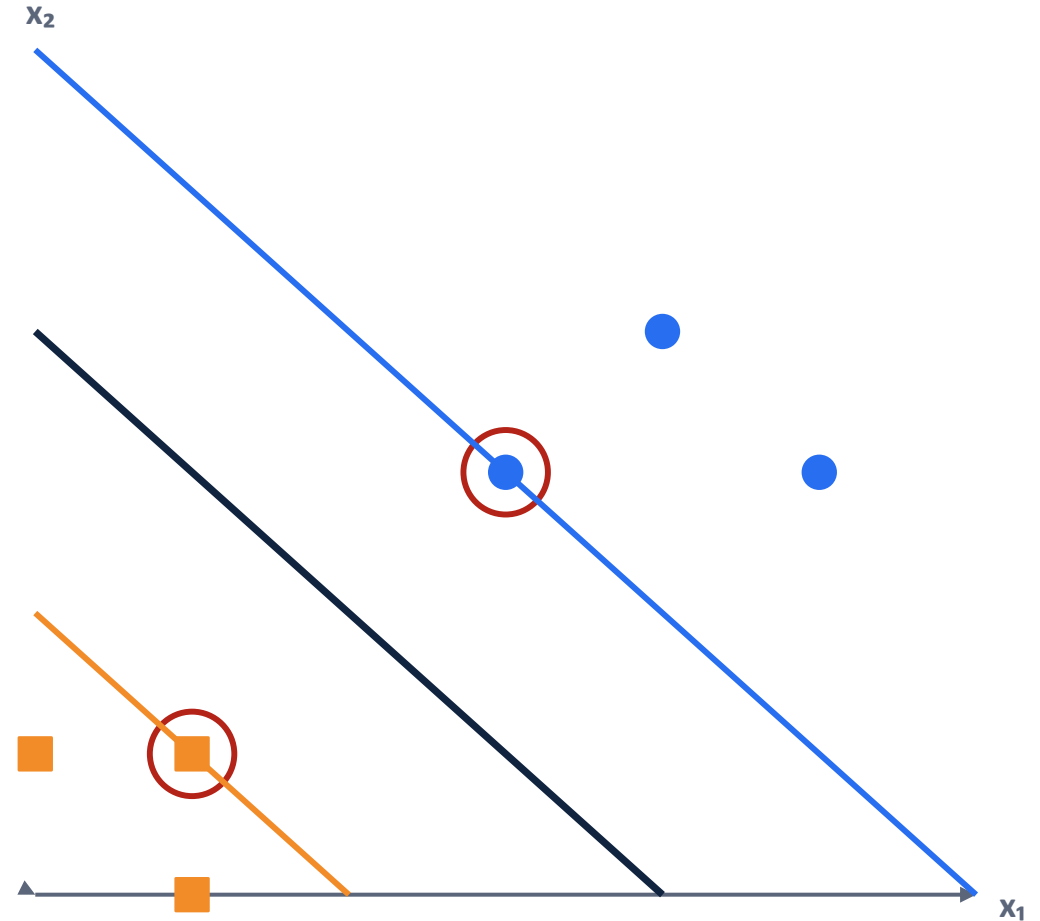
They alone define the hyperplane:

$$\mathbf{w} = \sum \alpha_i y_i \mathbf{x}_i$$

Only support vectors have $\alpha_i > 0$.
All other points usually have $\alpha_i = 0$.

Moving or removing far-away points usually changes nothing.

Moving a support vector can shift the boundary.



Support vectors are the hardest and most informative cases.

Final answer

Use this as the final slide or board summary.

Optimal hyperplane: $x_1 + x_2 - 4 = 0$

SVM-normalized form: $0.5x_1 + 0.5x_2 - 2 = 0$

Support vectors: $(3,3)$ and $(1,1)$

Margin boundaries: $x_1+x_2=6$ and $x_1+x_2=2$

Total margin: $2\sqrt{2} \approx 2.83$

Beginner idea: find the closest opposite-class points → put the boundary halfway → make it perpendicular.